

# Aadil Sood

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## SKILLS

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**Technical Skills:** Arduino, C, C++, Engineering Drawing, Fusion 360, Instrument Calibration, MATLAB, Metrology  
**Professional Skills:** Documentation, Leadership, Project Management, Teamwork, Technical Communication

## EXPERIENCE

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### Calibration Technician

July 2026 – August 2026

*Pylon Electronics - Wescan Calibration*

*Richmond, BC*

- Performed precision calibration and verification of dimensional, torque, and electronic measurement instruments
- Developed a Python-based interface to automate data collection from a Laser Scan Micrometer through RS-232 communication, reducing manual measurement entry for technicians
- Operated a wide range of precision metrology, including calipers, micrometers, weighing scales, torque analyzers, and digital multimeters (DMMs)
- Inspected, labeled, and certified calibrated equipment, adhering to quality assurance standards and laboratory protocols

## PROJECTS

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### 4-Bit Transistor-Based Binary Adder | *Falstad, BJT Circuits, Digital Logic*

Aug. 2026 – Present

- Designing a 4-bit binary adder entirely from discrete BJT transistors, without using integrated logic gates or logic gate ICs
- Building custom XOR, AND, and other logic gates from transistor networks to implement half and full adder circuits
- Using Falstad Circuit Simulator to prototype, test, and troubleshoot transistor-level logic circuits before physical implementation
- Applying Boolean logic, binary arithmetic, and carry propagation to cascade four 1-bit full adders into a 4-bit arithmetic unit

### Laser Scan Micrometer Automation | *Python, Data Automation, Serial Communication*

Feb. 2026 – June. 2026

- Developed a Python-based interface to automate Laser Scan Micrometer measurements and streamline the technician workflow
- Implemented serial communication to interface with the instrument and automate measurement data collection
- Troubleshoot communication and hardware issues to improve system reliability and ensure consistent operation
- Designed the workflow with future technicians in mind, reducing manual steps and simplifying instrument operation

### Automated Mechanical Claw Project | *C++, Arduino, Hand Tools*

Feb. 2026 – June. 2026

- Designed and fabricated a mechanical claw using sheet metal and hand tools
- Programmed an Arduino to control a servo motor based on input from an ultrasonic distance sensor
- Tested, refined, and optimized the system for consistent performance and reliability
- Implemented sensor-based logic to trigger automated actuation based on real-time distance thresholds

### Assistive Technology Toothpaste Roller | *Fusion 360, Assistive Design*

Nov. 2025 – Dec. 2025

- Designed an assistive toothpaste roller to support users with limited hand strength or dexterity, focusing on accessibility and ease of use
- Created detailed 3D CAD models (assemblies, tolerances, and moving components) using Fusion 360
- Delivered a formal design pitch presenting the problem statement, user needs, design rationale, and technical solution to instructors

### VEX Robotics Spin Up Robot | *C++, Fusion 360, Assembly Design*

Sept. 2022 – May 2023

- Designed, built, and tested a competitive robotics system to accomplish game objectives efficiently and reliably
- Used CAD and hands-on fabrication techniques to develop mechanical components and assemblies
- Integrated mechanical systems with sensors and control logic to optimize robot performance

## EDUCATION

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### Bachelor of Applied Science in Engineering Physics

*University of British Columbia*

Vancouver, BC

*Sept. 2025 – Present*